

II) I) $(P \vee Q) \rightarrow (Q \vee P)$

P	Q	$P \vee Q$	$Q \vee P$	$(P \vee Q) \rightarrow (Q \vee P)$
1	1	1	1	1
1	0	1	0	1
0	1	1	1	1
0	0	0	0	1

* $(P \vee Q) \rightarrow (Q \vee P)$ all values are "T" $\therefore$ This is propositional form is tautology (true)

II) $P \rightarrow ((P \vee Q) \vee R)$

P	Q	R	$P \vee Q$	$(P \vee Q) \vee R$	$P \rightarrow ((P \vee Q) \vee R)$
1	1	1	1	1	1
1	1	0	1	1	1
1	0	1	1	1	1
1	0	0	1	1	1
0	1	1	1	1	1
0	1	0	1	1	1
0	0	1	0	1	1
0	0	0	0	0	1

* $P \rightarrow ((P \vee Q) \vee R)$ all values are True, $\therefore$ This is propositional form is tautology

III) $P \rightarrow (Q \rightarrow (Q \rightarrow P))$

P	Q	$Q \rightarrow P$	$Q \rightarrow (Q \rightarrow P)$	$P \rightarrow (Q \rightarrow (Q \rightarrow P))$
1	1	1	1	1
1	0	1	1	1
0	1	0	0	1
0	0	1	1	1

* $P \rightarrow (Q \rightarrow (Q \rightarrow P))$ all values are True; $\therefore$ This is Propositional form is tautology.

iv) $((p \rightarrow q) \leftrightarrow q) \rightarrow p$

P	Q	$P \rightarrow Q$	$(P \rightarrow Q) \leftrightarrow Q$	$((P \rightarrow Q) \leftrightarrow Q) \rightarrow P$
1	1	1	1	1
1	0	0	0	1
0	1	1	1	0
0	0	1	0	0

* $((P \rightarrow Q) \leftrightarrow Q) \rightarrow P$ In here one value is false,
 $\therefore$ This is not propositional form is tautology.

v) $(P \wedge Q) \rightarrow (P \vee R)$

P	Q	$P \wedge Q$	$P \vee R$	$(P \wedge Q) \rightarrow (P \vee R)$
1	1	1	1	1
1	0	0	1	1
0	1	0	1	1
0	0	0	0	1

* $(P \wedge Q) \rightarrow (P \vee R)$ All values are true $\therefore$ This is propositional form is tautology

vi) $(p \rightarrow q) \leftrightarrow (q \rightarrow p)$

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \leftrightarrow (Q \rightarrow P)$
1	1	1	1	1
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1

* $(P \rightarrow Q) \leftrightarrow (Q \rightarrow P)$ In here two values are false,
 $\therefore$ This is not propositional form is tautology.

12.	P	Q	R	$Q \vee R$	$P \wedge Q$	$P \wedge (Q \vee R)$	$(P \wedge Q) \vee R$	$P \wedge (Q \vee R) \rightarrow (P \wedge Q) \vee R$
	1	1	1	1	1	1	1	1
	1	1	0	1	1	1	1	1
	1	0	1	1	0	0	1	1
	1	0	0	0	0	0	0	1
	0	1	1	1	0	0	1	1
	0	1	0	1	0	0	0	1
	0	0	1	1	0	0	1	1
	0	0	0	0	0	0	0	1

* $P \wedge (Q \vee R) \neq (P \wedge Q) \vee R$ (using Truth table)

* $P \wedge (Q \vee R) \rightarrow (P \wedge Q) \vee R$ is a tautology (∵ All values are True)

13.	P	Q	$\neg Q$	$\neg P$	$P \rightarrow Q$	$\neg Q \rightarrow (\neg P)$
	1	1	0	0	1	1
I)	1	0	1	0	0	0
	0	1	0	1	1	1
	0	0	1	1	1	1

$$P \rightarrow Q \equiv \neg Q \rightarrow (\neg P)$$

II)	P	Q	$\neg Q$	$P \rightarrow Q$	$\neg(P \rightarrow Q)$	$P \wedge (\neg Q)$
	1	1	0	1	0	0
	1	0	1	0	1	1
	0	1	0	1	0	0
	0	0	1	1	0	0

$$\neg(P \rightarrow Q) \equiv P \wedge (\neg Q)$$

III)	P	Q	$\neg P$	$P \rightarrow Q$	$\neg P \vee (\neg Q)$
	1	1	0	1	1
	1	0	0	0	0
	0	1	1	1	1
	0	0	1	1	1

$$P \rightarrow Q \equiv (\neg P) \vee Q$$

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IV)	P	Q	$\neg Q$	$P \rightarrow Q$	$P \wedge (\neg Q)$	0	$P \wedge (\neg Q) \rightarrow 0$
	1	1	0	1	0	0	1
	1	0	1	0	1	0	0
	0	1	0	1	0	0	1
	0	0	1	1	0	0	1

$$P \rightarrow Q \equiv P \wedge (\neg Q) \rightarrow 0$$

$$v) \quad P \rightarrow Q \equiv (\neg P) \vee Q \rightarrow (III)$$

$$P \leftrightarrow Q \equiv (P \rightarrow Q) \wedge (Q \rightarrow P)$$

using (III)

$$P \leftrightarrow Q \equiv ((\neg P) \vee Q) \wedge (Q \rightarrow P) \rightarrow (A)$$

using (FII)

$$Q \rightarrow P \equiv (\neg Q) \vee P$$

using (A)

$$P \leftrightarrow Q \equiv ((\neg P) \vee Q) \wedge ((\neg Q) \vee P) //$$

$$14) \quad P \rightarrow Q \equiv (\neg P) \vee Q \rightarrow (III)$$

$$r \rightarrow (P \rightarrow Q) \equiv (P \wedge r) \rightarrow Q$$

$$r \rightarrow (P \rightarrow Q) \equiv r \rightarrow (\neg P \vee Q) \quad (\because \text{using III})$$

$$\equiv \neg r \vee (\neg P \vee Q) \quad (\because \text{using III})$$

$$\equiv (\neg r \vee \neg P) \vee Q \quad (\text{Commutativity of conjunction})$$

$$\equiv \neg (r \wedge P) \vee Q \quad (\text{De Morgan's law})$$

$$\equiv \neg r \wedge \neg P \vee Q \rightarrow (1)$$

$$(P \wedge r) \rightarrow Q \equiv \neg (P \wedge r) \vee Q \quad (\because \text{using III})$$

$$\equiv \neg P \wedge \neg r \vee Q$$

$$\equiv \neg P \wedge \neg r \vee Q \quad (\because \text{commutativity of conjunction})$$

$$(1) = (2)$$

$$r \rightarrow (P \rightarrow Q) \equiv (P \wedge r) \rightarrow Q$$

			(I)			(II)
15.	p	q	$p \vee q$	$p \vee q$	$\neg(p \wedge q)$	$(p \vee q) \wedge \neg(p \wedge q)$
	1	1	0	1	0	0
	1	0	1	1	1	1
	0	1	1	1	1	1
	0	0	0	0	1	0

$$p \vee q \equiv (p \vee q) \wedge (\neg(p \wedge q)) //$$